Analysis of Post-CHF Heat Transfer for Downward Vertical Boiling in a Square Channel

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INTRODUCTION

The purpose of this work is to investigate post-critical heat flux (CHF) conditions for downward boiling flow in a square channel. This work modifies conditions and extend results found by McCord and Brooks [1], which considered both transient and stable film boiling phenomenon in the same system but with upwards flow. The value of performing an experiment like this is its potential to improve two-phase flows in nuclear reactors as a means of improved cooling. Minimal research has been done on downwards flow, particularly on post-CHF conditions, so this work seeks to fill the gaps in that area of the literature. Further objectives of this experiment are to streamline a method of measuring stable film boiling points and transient cooldown behavior for this system.

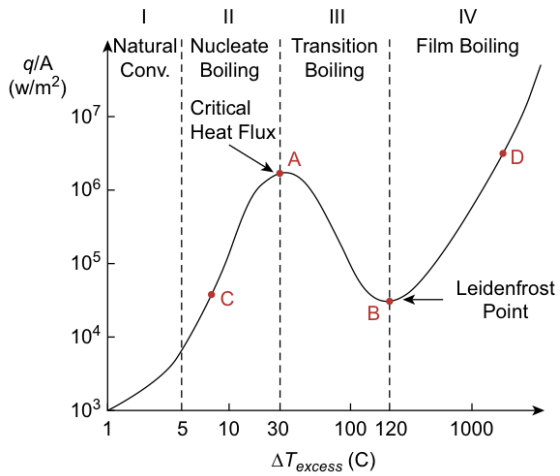


Fig. 1. Boiling curve as depicted by Forsberg [2].

As background, Figure 1 provides an ideal boiling curve, as drawn by Forsberg [2]. This figure is pertinent to the understanding of this work because it identifies what CHF should look like on the experimental plots. The post-CHF analysis this work seeks to perform focuses on the return from film boiling to transition boiling, known as rewet. This is not shown on a traditional boiling curve, but the regions remain the same and should be identifiable in our results.

METHODS

Figure 2 shows a schematic diagram of the facility used to conduct this experiment. The test section of Figure 1 indicates where the experimental data are collected, through several T-type thermocouples, including two at each the inlet (top of test section) and outlet (base of test section). The wall temperature data are collected by K-type thermocouples placed inside the wall of the heated channel, which are also used to calculate heat flux, as shown by Equation 2.

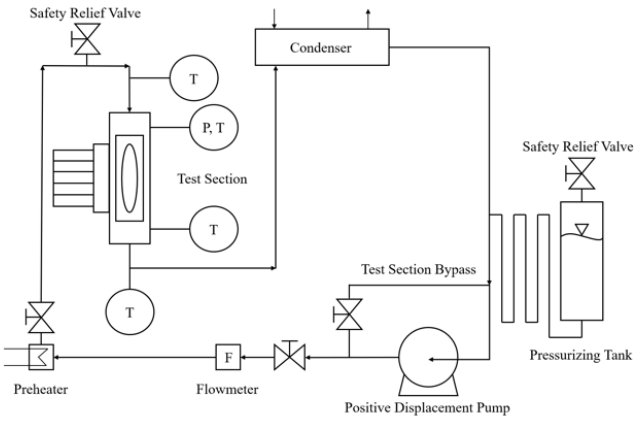


Fig. 2. Schematic of facility setup and test section.

During the experiment, water flows from the positive displacement pump to the preheater, through the test section, through the condenser (heat exchanger), and then back to the pump. There are several safety relief valves installed as this is a pressurized system. The test section is heated on one side by a copper heating element. The heat flux is changed by adjusting the supply of voltage to the block. The voltages are ramped up, held, ramped down, and held at various intervals throughout the experiment. This allows data to be collected at various points along the boiling curve, including stability points from film boiling to rewet. In addition to heat flux and temperature data collected by the thermocouples and shown in the Results section, we calculated the heat transfer coefficient of the test section over the course of the experiment using Equation 1,

$$h = \frac{q''}{\Delta T} \tag{1}$$

where q'' is the heat flux and ΔT is the temperature difference between the wall the and the fluid. The fluid temperature is calculated as an average of the inlet and outlet temperatures. Finally, the heat flux is calculated using a 1D heat conduction assumption shown in Equation 2 where k is the thermal conductivity of copper and ∇T is taken from the thermocouples inside the channel wall.

$$q'' = -k\nabla T \tag{2}$$

A complete description of the experimental facility, instrumentation, and data processing can be found in [2].

RESULTS

The results shown in this section are from a representative run of the experiment that demonstrates the features of the preliminary dataset.

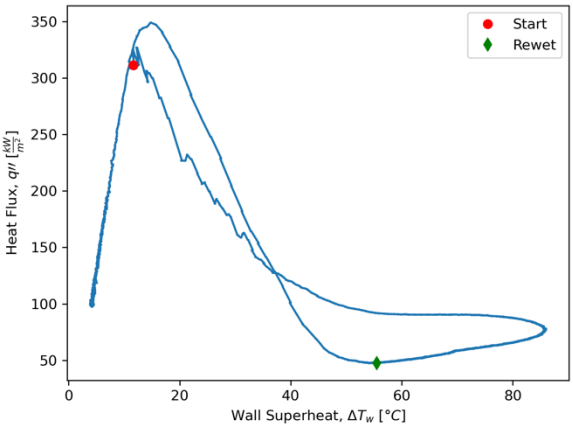


Fig. 3. Heat flux plotted over wall superheat for a preliminary run in the heated channel with downwards flow up to critical heat flux and back to rewet points.

As shown by Figure 3, we have included data collected from the facility starting just before CHF is reached. The red circle marker indicates where we started collecting the data; shortly thereafter, CHF is reached. The lower of the two peaks shown in Figure 3 indicates CHF, and the green diamond denotes rewet, indicating improved heat transfer due to the rapid return to nucleate boiling. The heat flux reaches its minimum at this point. Immediately after CHF, the steep, negatively sloped portion of the curve indicates transition boiling. The region between once the slope becomes nearly horizontal and prior to rewet indicates vapor film boiling. After rewet and up to the highest peak, the system returns to transition boiling, where liquid is again able to contact the surface. The heat flux spikes dramatically after rewet.

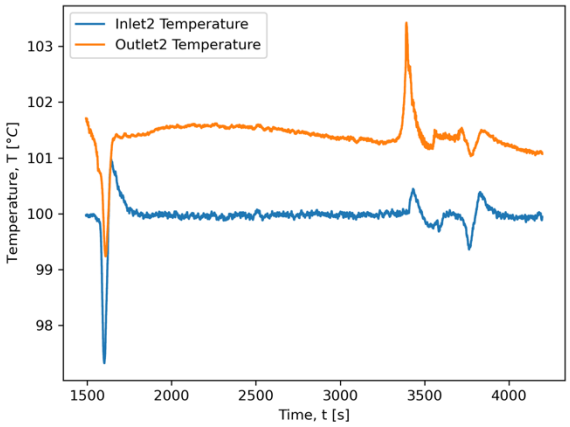


Fig. 4. Inlet and outlet temperatures over time. Inlet and outlet correspond to either end of the heated channel.

Figure 4 shows the inlet and outlet fluid temperatures of the test section over time in the experiment. The initial drop (~1600 s) in temperatures at both the inlet and the outlet is due to CHF, where heat transfer from the wall to the fluid drops dramatically. This drop correlates with a temperature spike shown in Figure 5, which depicts the wall temperature over time. At CHF, the wall temperature spikes because of the formation of a vapor blanket on the surface resulting in poor heat transfer to the fluid. The spike (~3400 s) in Figure 4 indicates rewet, corresponding to the drop in wall temperature in Figure 5. At rewet, the stable vapor film breaks down and the liquid contacts the surface of the wall, significantly improving heat transfer from the surface. This increases the temperature of the fluid and decreases the temperature of the wall.

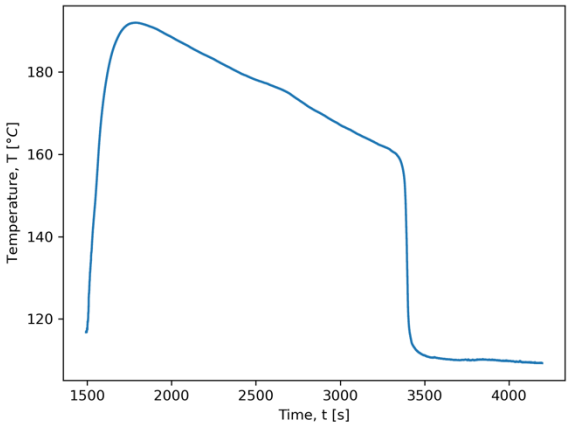


Fig. 5. Wall temperature of the heated channel plotted over time.

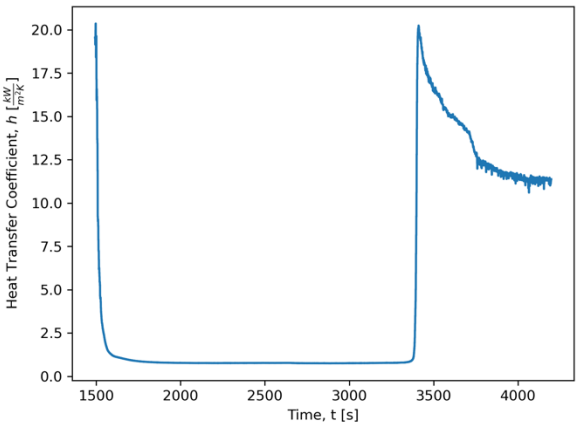


Fig. 6. Heat transfer coefficient calculated using Equation 1, over time in the heated channel.

Figure 6 shows the heat transfer coefficient, calculated using Equation 1, over time. As shown, after CHF, the heat transfer coefficient of the wall dramatically decreases because of a vapor blanket formation on the surface of the wall and the heat transfer is diminished. Once rewet occurs, we observe a spike in the heat transfer coefficient, as shown just before 3500 seconds in Figure 6. Between the CHF and rewet point, a stable vapor film boiling is recorded.

FUTURE WORK

Future work for this experiment includes an expansion of the conditions tested to compare with similar experiments, specifically analyzing the difference from past measurements with the facility in upward flow. Additionally, experimental data is being collected that spans a wide test matrix, including the effect of flow rate, pressure and liquid subcooling. The testing plan also includes reproducibility experiments to determine the sensitivity of the resulting features from fluctuations in system conditions.

REFERENCES

[1] M. McCord and C. Brooks, "Experimental comparison of the rewet phenomenon in transient and stable film boiling in low presure and low flow conditions," International Journal of Heat and Mass Transfer, 2023 (under review).

[2] C. H. Forsberg, "Chapter 12 - Special Topics," in *Heat Transfer Principles and Applications*, Academic Press, 2021.